

Sensory receptors and coding

The written notes for the sensory receptors and coding walkthrough. Week 5 competencies 8 to 12.

Step by step: where the signal travels

From touch to the brain, step by step

1. Pressure deforms the skin over a touch receptor.
2. Mechanically gated cation channels in the receptor ending open, Na⁺ enters, and the ending depolarizes. This graded change is the receptor potential.
3. The receptor potential spreads to the trigger zone. If it is above threshold there, voltage-gated Na⁺ channels open and the trigger zone fires action potentials.
4. A stronger stimulus gives a bigger receptor potential and a higher firing frequency, and it reaches the thresholds of more receptors.
5. The action potentials travel along the sensory axon, unchanged in size, through the peripheral nerve and the dorsal root to the spinal cord.
6. The signal climbs to the brain on its own pathway, a labeled line, and relays in the thalamus.
7. It reaches its own spot in the somatosensory cortex. Which line is active tells the brain the modality, and which spot is active tells it the location.

Lateral inhibition, step by step

1. A stimulus presses hardest at its center, so the receptor under the center fires fastest and its neighbors fire more slowly.
2. Each sensory neuron also drives inhibitory interneurons that reach its neighbors' pathways.
3. The fast-firing center neuron inhibits its neighbors more than they inhibit it.
4. The neighbors' signals drop much further than the center's.
5. The difference between center and edge grows, so the brain locates the stimulus more precisely.

Receptors and transduction

Every sensory pathway: stimulus, receptor, sensory neuron, integration in the CNS. Some signals reach conscious perception; many are acted on subconsciously. The nervous system can modulate the information at every synapse.

Kind of receptor	What it is	Examples
Simple neural receptor	A neuron with bare, free nerve endings	Pain and itch receptors
Complex neural receptor	Nerve ending wrapped in a connective tissue capsule	Pacinian corpuscle (vibration, touch)
Nonneural receptor	Modified epithelial cell that releases transmitter onto a sensory neuron	Hair cells of the ear

- Accessory structures help receptors without being receptors: the cornea and lens focus light; hairs on the skin detect air movement; the pinna funnels sound.
- Adequate stimulus: the form of energy a receptor responds to best. A strong enough stimulus of another kind can also activate it, as a blow to the eye makes you see light.
- Threshold: the minimum stimulus that activates a receptor. Some respond to one photon or one odorant molecule.
- Transduction opens or closes ion channels, directly or through a second messenger. Most receptors depolarize as cations enter; a few hyperpolarize as K⁺ leaves; light closes cation channels in photoreceptors.
- The receptor potential is a graded potential. In a neural receptor it can trigger action potentials at the trigger zone; in a nonneural receptor it changes transmitter release onto the sensory neuron.

Frequency coding

A stronger stimulus gives a bigger receptor potential and a higher firing frequency, up to a maximum, and more transmitter released at the axon terminal.

Population coding

Receptors have different thresholds, so a stronger stimulus recruits more of them. Many receptors together carry more information than one.

Receptor classes

Class	Adequate stimulus	Examples and locations
Chemoreceptor	Chemicals that bind the receptor	O ₂ , pH and glucose in the blood; taste buds; olfactory epithelium
Mechanoreceptor	Mechanical energy: pressure, stretch, vibration, gravity, acceleration, sound	Baroreceptors (pressure), osmoreceptors (cell stretch), touch receptors, muscle spindles, hair cells
Thermoreceptor	Temperature	Skin, hypothalamus
Photoreceptor	Light	Rods and cones of the retina
Nociceptor	Stimuli strong enough to damage tissue	Skin, joints, muscles, organs. Your textbook covers these with pain rather than as a fifth group

What the brain is told

Property	How it is coded
Modality	Which receptors are active and where their pathway ends in the cortex: labeled line coding. Qualities within a modality, such as red, green and blue, come from different receptors
Location	Which receptive fields are active; the cortex is a map with neighboring receptors processed in neighboring areas. Sound is located by the difference in arrival time at the two ears
Intensity	Frequency coding and population coding
Duration	How long the train of action potentials lasts, shaped by adaptation

- Stimulating the hand area of cortex during surgery is felt as a touch on the hand.
- One form of phantom limb pain comes from hyperactive secondary sensory neurons in the spinal cord; the brain places the pain in the missing limb.

- Most sensory pathways relay in the thalamus before the cortex. Smell does not. The midbrain receives visual information, the medulla receives input for sound and taste, and balance is processed mainly in the cerebellum.

Receptive fields and acuity

- Receptive field: the area that changes one sensory neuron's firing.
- Primary (first-order) sensory neuron synapses on a secondary (second-order) neuron in the CNS.
- Convergence: many primary neurons onto one secondary neuron, merging their fields into one large secondary receptive field; subthreshold stimuli can sum there.
- Arms and legs: much convergence, large fields; two points about 20 mm apart can feel like one. Fingertips: little convergence, as little as one to one; points a few millimeters apart feel like two.
- Lateral inhibition: the secondary neuron nearest the stimulus inhibits its neighbors, probably by opening K⁺ or Cl⁻ channels, increasing contrast so the stimulus is easier to locate. In vision it sharpens edges.

Adaptation and habituation

	Tonic (slowly adapting)	Phasic (rapidly adapting)
Response to a steady stimulus	Fires rapidly, then slows but keeps firing	Fires at onset, stops while the stimulus is constant, may fire at offset
Reports	The stimulus, continuously	Change
Examples	Baroreceptors, irritant receptors, some touch receptors, proprioceptors	Smell receptors; Pacinian corpuscles

- Mechanisms of adaptation: K⁺ channels open and repolarize the receptor, Na⁺ channels inactivate, or biochemical pathways change its responsiveness. Accessory structures can help: small middle ear muscles dampen loud sounds.
- After a phasic receptor adapts, only a stronger stimulus or removing the stimulus so the receptor resets produces a new signal.
- Perceptual threshold: the stimulus intensity needed for awareness. Habituation raises it by inhibitory modulation in secondary and higher neurons; attention can override it. Adaptation happens at the receptor; habituation happens in the CNS.